Syllabus Details

Syllabus ID:	12039
Syllabus Name:	**Database Systems_Các hệ cơ sở dữ liệu**
Course Name English:	**Database Systems**
Subject Code:	**DBI202**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 85' practical exam + 104h self-study
Pre-Requisite:	
Description:	- Knowledge about database systems has become an essential part of an education in computer science because database management has evolved from a specialized computer application to a central component of a modern computing environment. - The content of this course includes aspects of database management basic concepts, database design, database languages, and database-system implementation. Basing on these contents, the course emphasizes on how to organize, maintain and retrieve efficiently data and information from a DBMS.
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises given by instructor in class or at home and submit on time - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	Microsoft SQL Server
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1286/QĐ-ĐHFPT dated 11/22/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/22/2024

5 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	First Course in Database Systems	Jeffrey D. Ullman, Jennifer Widom	Pearson	2008	Third edition	9781292025827 or 9780136006374	True	True	False	
2	Database Management System	Raghu ramakrishnan	Mc Graw-Hill		Third edition	978-8131769591	False	True	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Database Management System (DBMS): A Practical Approach	Rajiv Chopra	SChand Publications			9788121932455	False	True	False	
4	Course slides (.pptx)						False	False	False	
5	Labs & assignment						False	False	False	

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand thedatabase concepts and database management system software
2	CLO2	Understand the relation model of data and Algebraic Query Language
3	CLO3	Understand data normalization and apply normalization techniques in database design
4	CLO4	Be able to model an application's data requirements using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model.
5	CLO5	Be proficient in structure query language including Data Definition Language(DDL) and Data Manipulation Language(DML)
6	CLO6	Understand PL/SQL concepts and manipulate with View, Cursors, Stored Procedures, Functions, Database Triggers
7	CLO7	Apply the Index in database design and query optimization

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Chapter 1. The Worlds of Database Systems 1.1 The Evolution of Database Systems 1.2 Overview of Database Management System	Offline	LO1	I	Textbook, slides	DBI202	Read chapter 1 in text book, focus on 1.1 and 1.2	
2	1.3 Outline of Database-System Studies Assignment Introduction (individual)	Offline	LO1	I	Textbook, slides	DBI202	Read chapter 1 in text book, focus on 1.1 and 1.2	
3	Chapter 2. The Relational Model of Data 2.1 An Overview of Data Models	Offline	LO2	IT	Textbook, slides		Read chapter 2 in text book, focus on 2.1 and 2.2	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
4	2.2 Basics of the Relational Model	Offline	LO2	I,T	Textbook, slides		Read chapter 2 in text book, focus on 2.1 and 2.2	
5	2.4 An Algebraic Query Language	Offline	LO2	I,T	Textbook, slides		Read chapter 2 in text book, focus on 2.4	
6	2.4 An Algebraic Query Language (cont.)	Offline	LO2	I,T	Textbook, slides		Read chapter 2 in text book, focus on 2.4	
7	Lab1 assistance	Offline	LO1, LO2	U	Textbook, slides, lab's questions		Do lab 1 as homework	
8	Lab1 assistance (cont.)	Offline	LO1, LO2	U	Textbook, slides, lab's questions		Do lab 1 as homework	
9	Chapter 3. Design Theory for Relational Databases 3.1 Functional Dependencies	Offline	LO3	I, T	Textbook, slides		Read chapter 3 in text book, focus on 3.1 and 3.2	
10	3.2 Rules About Functional Dependencies	Offline	LO3	I, T	Textbook, slides		Read chapter 3 in text book, focus on 3.1 and 3.2	
11	3.3 Design of Relational Database Schema	Offline	LO3	I, T	Textbook, slides		Read chapter 3 in text book, focus on 3.3	
12	3.3 Design of Relational Database Schema (cont.)	Offline	LO3	I, T	Textbook, slides		Read chapter 3 in text book, focus on 3.3	
13	3.5 Normal Forms (1NF, 2NF)	Offline	LO3	I, T	Textbook, slides		Read chapter 3 in text book, focus on 3.5	
14	3.5 Normal Forms (cont. 3NF, BCNF)	Offline	LO3	I, T	Textbook, slides		Read chapter 3 in text book, focus on 3.5	
15	Lab 2 assistance	Offline	LO3	U	'Textbook, slides, lab's questions		Do lab 2 as homework	
16	Lab 2 assistance (cont.)	Offline	LO3	U	'Textbook, slides, lab's questions		Do lab 2 as homework	
17	Progress test 1 (<=30') Assignment assistance	Offline	LO1, LO2, LO3	I, U	Textbook, slides,assignment's questions		Do assignment as homework	
18	Assignment assistance (cont.)	Offline	LO1, LO2, LO3	I, U	Textbook, slides,assignment's questions		Do assignment as homework	
19	Chapter 4. High-Level Database Models 4.1 The Entity / Relationship Model	Offline	LO4	I, T	Textbook, slides		Read chapter 4 in text book, focus on 4.1, 4.2 and 4.3	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
20	4.2 Design Principles 4.3 Constraints in the E / R Model	Offline	LO4	I, T	Textbook, slides		Read chapter 4 in text book, focus on 4.1, 4.2 and 4.3	
21	4.4 Weak Entity Sets 4.5 From E / R Diagrams to Relational Models	Offline	LO4	I, T, U	Textbook, slides		Read chapter 4 in text book, focus on 4.4, 4.5 and 4.6	
22	4.6 Converting Subclass Structures to Relations	Offline	LO4	I, T, U	Textbook, slides		Read chapter 4 in text book, focus on 4.4, 4.5 and 4.6	
23	Lab 3 assistance	Offline	LO4	U	Textbook, slides, lab's questions		Do lab 3 as homework	
24	Lab 3 assistance (cont.)	Offline	LO4	U	Textbook, slides, lab's questions		Do lab 3 as homework	
25	Assignment assistance	Offline	LO1, LO2, LO3, LO4	U	Textbook, slides,assignment's questions		Do assignment as homework	
26	Assignment assistance (cont.)	Offline	LO1, LO2, LO3, LO4	U	Textbook, slides,assignment's questions		Do assignment as homework	
27	Chapter 6. The Database Language SQL 6.1 Data Definition Lanaguage (DDL)	Offline	LO5	I,T	Textbook, slides		Read chapter 6 in text book, focus on 6.1 and 6.2	
28	DDL (cont.)	Offline	LO5	I,T	Textbook, slides		Read chapter 6 in text book, focus on 6.1 and 6.2	
29	6.2. Implement constraints on attributes with MS SQL Server (Keys and Foreign Keys, UNIQUE, CHECK,..)	Offline	LO5	I,T	Textbook, slides		Read chapter 7 in text book, focus on 7.1 and 7.2	
30	6.2. Implement constraints on attributes with MS SQL Server (cont.)	Offline	LO5	I,T	Textbook, slides		Read chapter 7 in text book, focus on 7.1 and 7.2	
31	6.3 DML introduction & Basic of SQL Queries	Offline	LO5	I,T	Textbook, slides		Read chapter 6 in text book	
32	DML (cont.)	Offline	LO5	I,T	Textbook, slides		Read chapter 6 in text book	
33	6.4 Query on more one relation	Offline	LO5	I,T	Textbook, slides		Read chapter 6 in text book	
34	6.4 Query on more one relation (cont.)	Offline	LO5	I,T	Textbook, slides		Read chapter 6 in text book	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
35	6.5 Nested Queries in SQL	Offline	LO5	T,U	Textbook, slides		Read chapter 6 in text book	
36	6.5 Nested Queries in SQL (cont.)	Offline	LO5	T,U	Textbook, slides		Read chapter 6 in text book	
37	6.6 Aggregation Queries in SQL	Offline	LO5	T,U	Textbook, slides		Read chapter 6 in text book	
38	6.6 Aggregation Queries in SQL (cont.)	Offline	LO5	T,U	Textbook, slides		Read chapter 6 in text book	
39	6.7 Database Modifications (INSERT, UPDATE statement)	Offline	LO5	T,U	Textbook, slides		Read chapter 6 in text book	
40	6.7 Database Modifications (cont. DELETE statement)	Offline	LO5	T,U	Textbook, slides		Read chapter 6 in text book	
41	Chapter 7. Practical Issues of database application 7.1. Index	Offline	LO7	I,T, U	Textbook, slides		Read chapter 6 in text book	
42	Index (cont.)	Offline	LO7	I,T, U	Textbook, slides		Read chapter 6 in text book	
43	7.2. Transaction, View and Query Optimization	Offline	LO6, LO7	I,T, U	Textbook, slides		Read chapter 6 in text book	
44	7.2. Transaction, View and Query Optimization (cont.)	Offline	LO6, LO7	I,T, U	Textbook, slides		Read chapter 6 in text book	
45	Lab 4 assistance	Offline	LO5, LO6, LO7	U	Textbook, slides, lab's questions		Do lab 4 as homework	
46	Lab 4 assistance (cont.)	Offline	LO5, LO6, LO7	U	Textbook, slides, lab's questions		Do lab 4 as homework	
47	Assignment assistance (focus on query)	Offline	LO1, LO2, LO3, LO4, LO5, LO7	U	'Textbook, slides,assignment's questions		Do assignment as homework	
48	Assignment assistance (focus on query) (cont.)	Offline	LO1, LO2, LO3, LO4, LO5, LO7	U	'Textbook, slides,assignment's questions		Do assignment as homework	
49	Chapter 8. Constraints and T-SQL Programming 8.1 Triggers	Offline	LO6	T,U	Textbook, slides		Read chapter 7 in text book, focus on 7.5	
50	8.2 Constraint with triggers	Offline	LO6	T,U	Textbook, slides		Read chapter 7 in text book, focus on 7.5	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
51	8.3 View, Function	Offline	LO6	T	Textbook, slides		Read chapter 7 in text book	
52	8.3 View, Function (cont.)	Offline	LO6	T	Textbook, slides		Read chapter 7 in text book	
53	8.4 Cursors	Offline	LO6	T	Textbook, slides		Read chapter 9 in text book	
54	8.5 Implement stored procedure with MS SQL Server	Offline	LO6	T	Textbook, slides		Read chapter 9 in text book	
55	Progress test 2 Lab 5 assistance	Offline	LO4, LO5, LO6, LO7	U	Textbook, slides, lab's questions		Do lab 5 as homework	
56	Lab 5 assistance (cont.)	Offline	LO4, LO5, LO6, LO7	U	Textbook, slides, lab's questions		Do lab 5 as homework	
57	Assignment review	Offline	LO1, LO2, LO3, LO4, LO5, LO6, LO7	U	'Textbook, slides,assignment's questions		Do assignment as homework	
58	Assignment review (cont.)	Offline	LO1, LO2, LO3, LO4, LO5, LO6, LO7	U	'Textbook, slides,assignment's questions		Do assignment as homework	
59	Review	Offline	All LOs					
60	Review	Offline	All LOs					

30 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1.1	Why do we need database? How many stages of upgrade or development has the database gone through? Why did that happen? What is Database System?
2	2	CQ1.2	What is a Database System and the role of Database Systems? What is a Database Management System? What and how can programmers do with the database?
3	3	CQ1.3	What is Data Model? What belong to a Data Model?How can we describe and give some examples of the Relations, Attributes, Tuples, Domains?
4	4	CQ2.1	What are Relation instances, Schema, DB schema. How can we use SQL to define relation schemas?
5	5	CQ2.2	What is an Algebra? What is Relational Algebra? Why do we need Algebraic Query Language?

No.	Session No	Name	Details
6	6	CQ2.3	What are usecases of Relational Algebra Operations?
7	7	CQ3.1	What are constraints on Relations?
8	8	CQ3.2	How can we express constrains by relational algebra?
9	9	CQ3.3	How can we use relational algebra to define constrains in some database?
10	13	CQ5.1	What are Functional Dependencies? Why you need rules about Functional Dependencies? What is usecase of each rule?
11	14	CQ5.2	What are keys, super-keys, closure-sets, closing sets of FD's and how to determine them?
12	15	CQ5.3	What are Anomalies? Can we solve Anomalies? How to solve?
13	16	CQ6.1	Why should we decompose a relation?
14	17	CQ6.2	What is a simple way to normalize schema to 1NF, 2NF, 3NF, BCNF? Please, give some examples.
15	18	CQ6.3	What are the good, bad and ugly of decomposition?
16	22	CQ8.1	What is Database model? How can we describe the Database Design Process?
17	23	CQ8.2	How many Level Database Model? What are they?
18	24	CQ8.3	Why do we use Conceptual Design (or Logical Design)?
19	25	CQ9.1	What are Entity, Entity Set, Entity Instances, Attributes and others in ERD?
20	26	CQ9.2	What is a relationship and how to realize it? How can we draw an ERD?
21	27	CQ9.3	Are there any principles or constraints in designing ERDs?
22	34	CQ12.1	How can we use simple queries in some examples?
23	35	CQ12.2	How can we query data involving more than one relation? Please, give some examples.
24	36	CQ12.3	What are subqueries? Can you apply them im some examples?
25	37	CQ13.1	How can we use grouping and aggregate functions? Please, give some examples.
26	38	CQ13.2	What is HAVING clause? When should we use grouping and aggregate functions with HAVING clauses? Why?
27	39	CQ13.3	What are DDL, DML? How can we describe INSERT, UPDATE, DELETE queries? Please, give some examples.
28	46	CQ16.1	How can we use SQL to create/modify constraints of a relation?
29	47	CQ16.2	What is a trigger? Why do we use triggers?
30	48	CQ16.3	How many components in a trigger? How to create a trigger?

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	1	20.0%	>0	28 slots			N/A	All subjects in syllabus	in class, by teacher	
2	Lab	on-going	5	10.0%	>0	90'			N/A		in class, by teacher	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
3	Practical Exam	on-going	1	30.0%	>0	85'			N/A		By exam board	
4	Progress test	on-going	2	10.0%	>0	30'		Multiple choices; Marked by Computer or a suitable format	20		in class, by LMS system	Instruction and schedules for Progress Tests must be presented in the Course Implementation Plan approved by director of the campus. Progress test must be taken right after the last lectures of required material. Instructor has responsibility to review the test for students after graded. Progress test must be taken right after the last lectures of required material. Instructor has resposibility to review the test for students after graded.
5	Final exam	Final exam	1	30.0%	4	60			50	All subjects in syllabus	Exam room	The exam questions must be updated or different at least 70% to the previous ones.